



SAHYATRI

AN INTEGRATED SOLUTION FOR SAFER AND SMARTER TRANSPORTATION

Project Overview

The **Sahyatra** is an integrated platform designed to enhance road safety, improve transportation efficiency, and assist in emergency scenarios. The system leverages cutting-edge technologies such as real-time processing, AI-powered detection, and external API integrations to create a safer and smarter transportation environment for users.

Problem Statement

Road accidents and driver fatigue contribute significantly to fatalities worldwide.

- ❑ Lack of real-time information about traffic, weather, and emergencies hinders road safety.
- ❑ Inefficient transport systems discourage public transport usage, adding to congestion and environmental impact.

Objectives and Goals

- Enhance Road Safety:**

- Detect drowsiness in drivers and provide timely alerts.
- Enable real-time accident reporting and emergency responses.

- Improve Transportation Efficiency:**

- Offer features like metro route planning and traffic updates.
- Provide weather-based route recommendations.

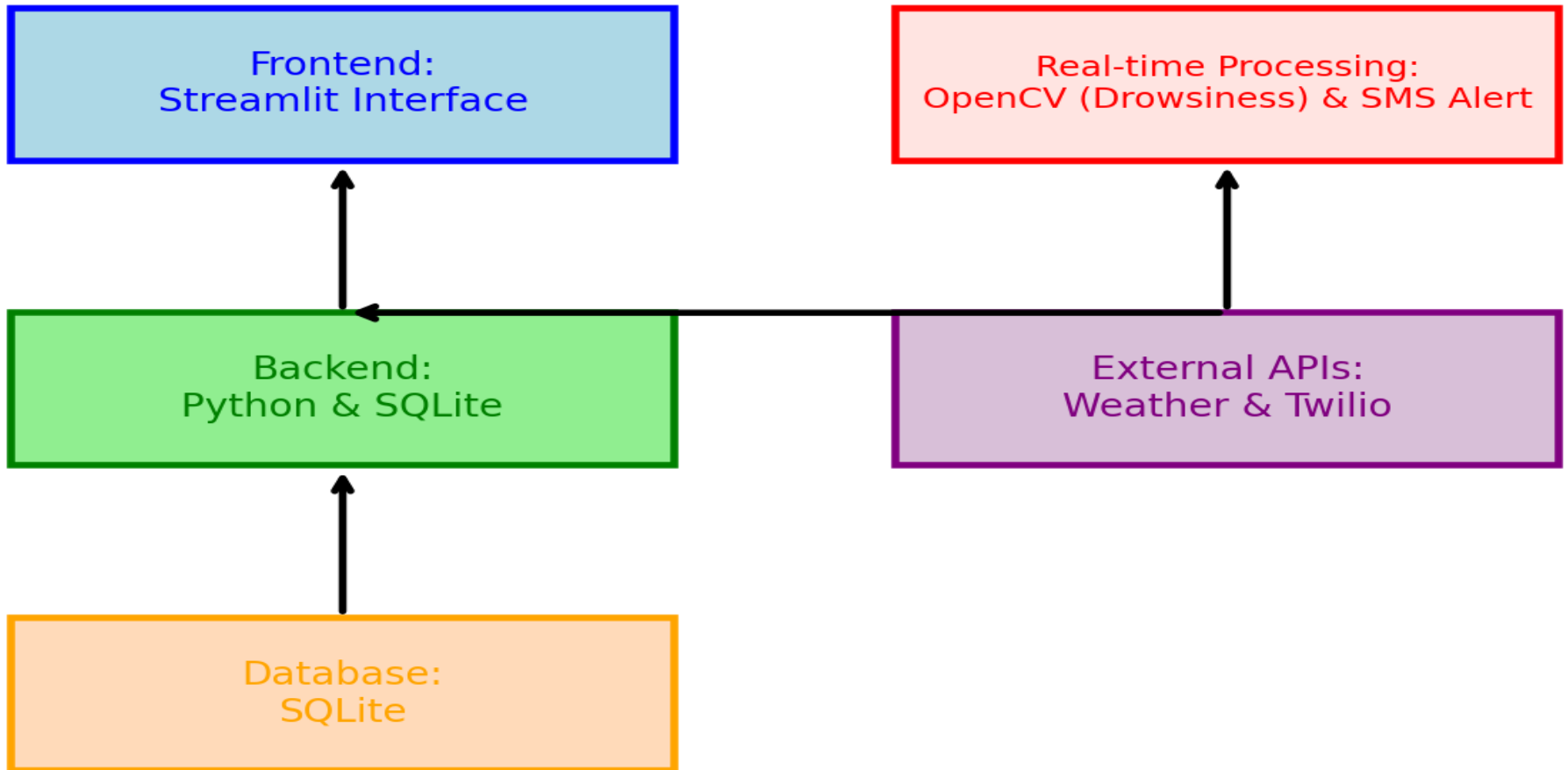
- Enable Proactive Emergency Response:**

- Integrate SOS functionality for immediate assistance.
- Notify authorities and emergency contacts through automated alerts.

- Promote Public Awareness:**

- Offer real-time updates on weather and traffic.
- Share safety tips to reduce risks during unfavorable conditions.

System Architecture



Ace Features

Real -Time Drowsiness Analyzer and Alarm System

- Detects driver fatigue using a webcam.
- Alerts the driver with a sound alarm if drowsiness is detected.

Emergency Response System

- Allows users to request emergency assistance.
- Sends SMS notifications using Twilio.
- Logs emergency requests in a database for tracking.

•Weather Monitoring

- Provides real-time weather updates.
- Displays weather alerts and safety tips.

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FUTURE ASPECTS

1. Integration of AI/ML for Advanced Analytics
2. Mobile Application Development
3. IoT Integration
4. Advanced Emergency Response
6. Crowd-Sourced Data Collection
5. Integration of Additional Transport Modes
6. Advanced Security Measures
7. Smart City Integration
8. Environmental Impact Monitoring
9. Collaboration with Authorities